



Legumes, the much maligned superfood

John Livesey PhD
Scientific Officer
Department of Endocrinology
Christchurch Hospital
New Zealand

Legumes have long been maligned as a flatulence-inducing food of the poor, but modern research has elevated the humble legume to a superfood and damned the animal products preferred by the more affluent.

A key characteristic of legumes is that they all have the ability to fix atmospheric nitrogen. Most plants can only use nitrogenous compounds in the soil to make their proteins, but legumes use the nitrogen in the air as well. This means that legumes are ecological good guys; they require less fertilizer when grown for food and they can be grown as a green manure to enrich the soil with nitrogenous material for use by other plants. These virtues are becoming very relevant as nitrogenous chemical fertilizer prices are currently rising rapidly as the cost of energy increases.

When scientific analysis of food became possible a hundred and fifty years ago, it became apparent that legumes were a nutritious food, being high in protein and minerals. For example, per 1000 calories, broad beans contain almost as much protein as a t-bone steak (77 grams compared to 87) but twice as much iron and six times as much calcium. They also have only 6% of the fat, and what fat there is in beans is healthier, as chemically, it is less saturated. In addition, legumes are excellent sources of fibre whereas all animal products contain none at all.

Excitingly though for vegetarians and vegans, epidemiological research has more recently elevated legumes from being simply a nutritious food to being a super food. One such research project was the important Seven Countries Study, the results of which lead to the popularisation of the Mediterranean diet. This study began in 1958 and collected dietary and medical data from 13,000 middle-aged men living in The Netherlands, Finland, Italy, Yugoslavia, Greece, Japan and the USA. As the investigators were particularly interested in the relationship between diet and heart disease, one of the ways they analysed the data they collected was to correlate the amount of different types of foods consumed with the death rate from coronary heart disease. They found that the food that was most strongly associated ($r = -0.82$) with a reduction in the death rate was legumes.

Incidentally, in the same study, oil and moderate alcohol consumption too were associated ($r = -0.57$ and -0.61) with a reduction in death rate; while butter ($r = +0.89$), pastries ($r = +0.75$), meat ($r = +0.65$), sugar and milk (both $r = +0.60$) were all associated with an increase in the death rate from heart disease. The lead investigator in the study, Dr Ancel Keys, was so impressed by the findings that he adopted a Mediterranean diet, retired to Italy, with his wife published a book of bean recipes, and lived to 100.

Further evidence for the remarkable health benefits of legumes came from the Californian Adventist Health Study where Dr Gary Fraser, an ex-patriate Kiwi, found that non-vegetarians who rarely ate legumes had 2.5 times the risk of getting colon cancer that vegetarians had. However, if the non-vegetarians consumed legumes more than twice a week, this meat-induced rise in the colon cancer rate was completely reversed.

More recently, an Australian-lead international research group examined the diet and death rates of 785 people over 70 years of age in five study groups in four countries

Legumes, the much maligned superfood

(Japanese in Japan, Swedes in Sweden, Anglo-Celtic in Australia, Greeks in Australia and Greece). Their dietary intake was categorised into nine food groups, and of these, only legumes were statistically significantly associated with a reduced death rate. Every 20 gram per day increase in legume consumption was associated with a reduction of 8% in the death rate.

The popular book, *The Blue Zones*, chronicles the visits of Dan Buettner and his team to areas of the world where the inhabitants are particularly long-lived. The four Blue Zones described in the book are in the mountains of Sardinia, in Okinawa, the Seventh Day Adventists in California, and in the Nicoya Peninsula, Costa Rica. In these travels Dan found that that "Beans are a cornerstone of the Blue Zone diets". A further Blue Zone has since been identified on the Greek island of Ikaria.

Well known foods made primarily from legumes include falafel, hummus, tofu, tempeh, peanut butter, baked beans, bean sprouts and soy milk. All these can be readily purchased in New Zealand and, apart from tempeh, can be made at home if you have the time. Also available frozen from Asian food specialists are edamame (green soy beans in their pods) and natto (fermented soy beans).

There is though one possible problem with preparing bean-based foods from scratch at home. Many of the cheaper dried beans imported into New Zealand have been heat-treated to meet biosecurity requirements. This results in beans that won't sprout and will remain fairly firm even after prolonged cooking. It is possible, but more expensive, to import non-heat treated beans, so check carefully with your supplier if these are what you require.

Natto is a particularly interesting leguminous food because it is both the best dietary source of vitamin K2, and the only vegan source of K2 (it also has a curious sticky/stringy texture). Most Westerners get their K2 from the small amounts in animal products, particularly cheese. Thus natto, being high in protein, fat (oil) and K2, and being strongly flavoured, is the real vegan 'cheese'.

There is increasing interest in vitamin K2 because it is metabolised much more slowly than vitamin K1 (the sort found in green leafy vegetables) and thus may be more effective at promoting bone and cardiovascular health. It has the doubly desirable (and paradoxical) actions of increasing calcification of bone but decreasing calcification of arteries. Indeed, a recent Dutch study found that higher intakes of vitamin K2, but not K1, were associated with lower rates of death both from heart disease and death from any cause. In addition, there is evidence related to bone metabolism which also suggests that most people do not get enough of the K vitamins, hence natto should probably be eaten more frequently in New Zealand. Note though that natto should be eaten cold and should not be cooked, as heating destroys the K2. I find it goes well with salads.

And my favourite legume recipe is the lentil and red pepper stew in Rose Elliot's *Vegetarian Dishes of the World*. Here it is:

Lentil and Red Pepper Stew

- 125 g red lentils
- 125 g haricot beans
- 2 large onions
- 700 g red peppers
- 40 g olive oil
- 1 litre stock or water
- 4 tablespoons tomato purée
- Salt and sugar to taste

Soak lentils and beans overnight, rinse and drain. Chop the onions and slice the peppers, discarding the cores and seeds. Fry the onion in the oil for 10 minutes, add the peppers and fry for a further 5 minutes. Add the lentils and beans, and the water or stock. Bring to boil and simmer gently with lid half on saucepan until the beans are tender (1 - 1¼ hours). Mix in tomato purée, add salt and sugar if desired. If necessary, thicken by boiling and stirring with lid off. Serves 4.

REFERENCES

Buettner, D. *The Blue Zones: Lessons for Living Longer From the People Who've Lived the Longest*, Washington, National Geographic.

Darmadi-Blackberry, I. *et al.* Legumes: the most important dietary predictor of survival in older people of different ethnicities. *Asia Pacific J Clin Nutr* 2004;13:217.

Legumes, the much maligned superfood

Elliot, R. (1996) *Vegetarian Dishes of the World*, London, Thorsons.

Fraser, GE. (2003) *Diet, Life Expectancy, and Chronic Disease*. Oxford, OUP.

Keys, A and M. (1967) *The Benevolent Bean*. New York, Farrar, Straus and Gioux.

Porphyry's People [home page](#).

NZT: 14/Jul/08 21:55:02. GMT: 14/Jul/08 09:55:02.